NLP POS tagging exercises

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Q1 Consider the following tagged corpus:

- The | DET book | NOUN is | AUX at | ADP the | DET bank | NOUN
- I | PRON will | AUX book | VERB and | CCONJ fly | VERB
- I | PRON will | AUX bank | VERB on | ADP the | DET game | NOUN and | CCONJ game | VERB
- The | DET fly | NOUN is | AUX on | ADP the | DET game | NOUN book | NOUN

Q1.1 Build the transition and emission probability tables for the following tagged corpus, by filling table 1 and table 2. In table 1, the rows correspond to the first tag of the bigram (t_{i-1}) and the columns correspond to the second tag (t_i). In table 2, the rows correspond to the tag t_i and the columns to the words w_i. To make the exercise slightly less tedious, I already filled in the 0 values.

$t_{i-1} \backslash t_i$	DET	NOUN	AUX	ADP	PRON	VERB	CCONJ	</s>
<s>		0	0	0		0	0	0
DET	0		0	0	0	0	0	0
NOUN	0			0	0	0		
AUX	0	0	0		0		0	0
ADP		0	0	0	0	0	0	0
PRON	0	0		0	0	0	0	0
VERB	0	0	0		0	0		
CCONJ	0	0	0	0	0		0	0

Table 1: Transition probability table to be filled

$t_i \backslash w_i$	the	book	is	at	bank	I	will	and	fly	on	game
DET		0	0	0	0	0	0	0	0	0	0
NOUN	0		0	0		0	0	0		0	
AUX	0	0		0	0	0		0	0	0	0
ADP	0	0	0		0	0	0	0	0		0
PRON	0	0	0	0	0		0	0	0	0	0
VERB	0		0	0		0	0	0		0	
CCONJ	0	0	0	0	0	0	0		0	0	0

Table 2: Emission probability table to be filled

🔑 **Q1.1** See table 3 and table 4.

$t_{i-1} \backslash t_i$	DET	NOUN	AUX	ADP	PRON	VERB	CCONJ	</s>
<s>	2/4	0	0	0	2/4	0	0	0
DET	0	5/5	0	0	0	0	0	0
NOUN	0	1/6	2/6	0	0	0	1/6	2/6
AUX	0	0	0	2/4	0	2/4	0	0
ADP	3/3	0	0	0	0	0	0	0
PRON	0	0	2/2	0	0	0	0	0
VERB	0	0	0	1/4	0	0	1/4	2/4
CCONJ	0	0	0	0	0	2/2	0	0

Table 3: Transition probability table

$t_i \backslash w_i$	the	book	is	at	bank	I	will	and	fly	on	game
DET	5/5	0	0	0	0	0	0	0	0	0	0
NOUN	0	2/6	0	0	1/6	0	0	0	1/6	0	2/6
AUX	0	0	2/4	0	0	0	2/4	0	0	0	0
ADP	0	0	0	1/3	0	0	0	0	0	2/3	0
PRON	0	0	0	0	0	2/2	0	0	0	0	0
VERB	0	1/4	0	0	1/4	0	0	0	1/4	0	1/4
CCONJ	0	0	0	0	0	0	0	2/2	0	0	0

Table 4: Emission probability table

Q1.2 Now tag the following sentences using your HMM model and viterbi:

- The game is on
- I book the game

🔑 **Q1.2** *The game is on* is tagged as: DET NOUN AUX ADP, see Viterbi chart in table 5.

	The	game	is	on
DET	**0.50000**	0	0	0
NOUN	0	**0.16667**	0	0
AUX	0	0	**0.02778**	0
ADP	0	0	0	**0.00926**
PRON	0	0	0	0
VERB	0	0	0	0
CCONJ	0	0	0	0

Table 5: Viterbi chart for *the game is on*

I book the game cannot be tagged with our model, all paths lead to 0 probabilities from the 2nd column. See table 6. We also need smoothing!

	I	book	the	game
DET	0	0		
NOUN	0	0		
AUX	0	0		
ADP	0	0		
PRON	**0.5**	0		
VERB	0	0		
CCONJ	0	0		

Table 6: Viterbi chart for *I book the game*